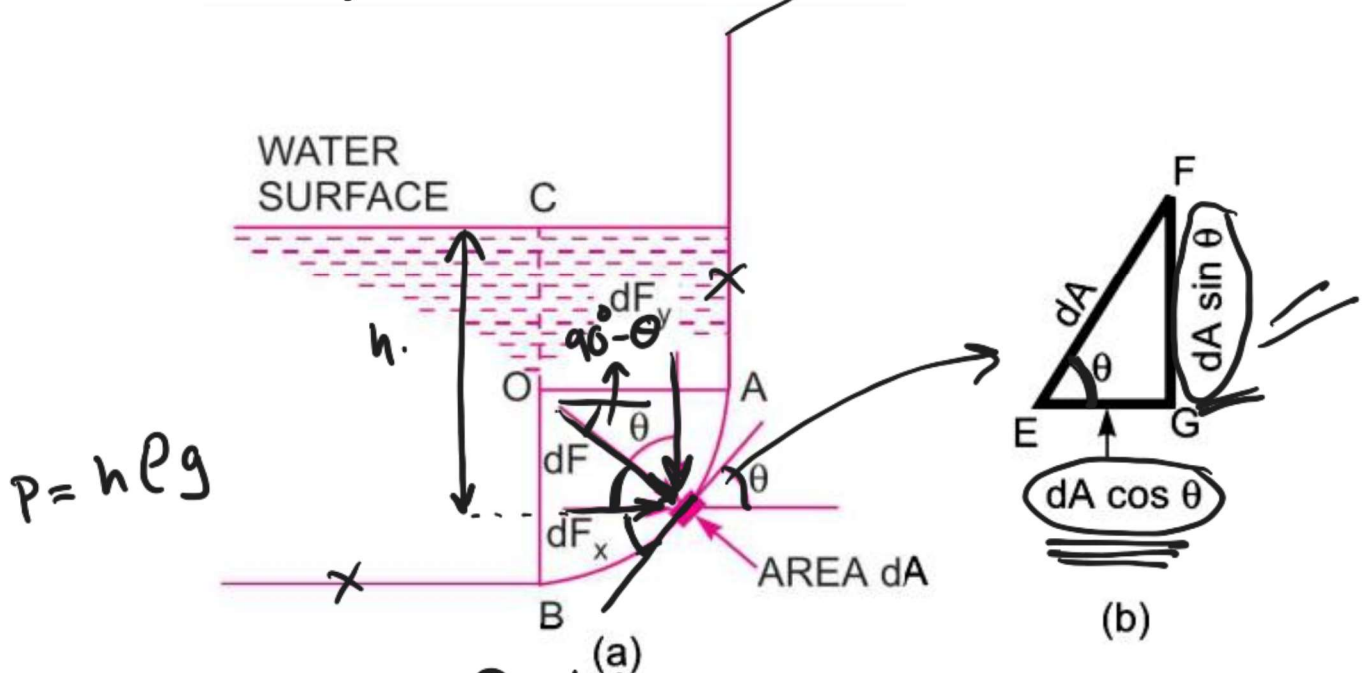


## Hydrostatic Forces on Curved Surfaces



$$p = h \rho g$$

$$dF = p dA = h \rho g dA$$

$$dF_x = dF \cos(90 - \theta) = dF \sin \theta$$

$$= h \rho g \sin \theta dA$$

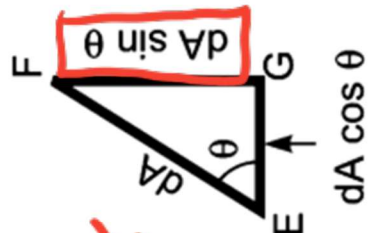
$$dF_y = dF \cos \theta = h \rho g \cos \theta dA$$

$$F_x = \int dF_x$$

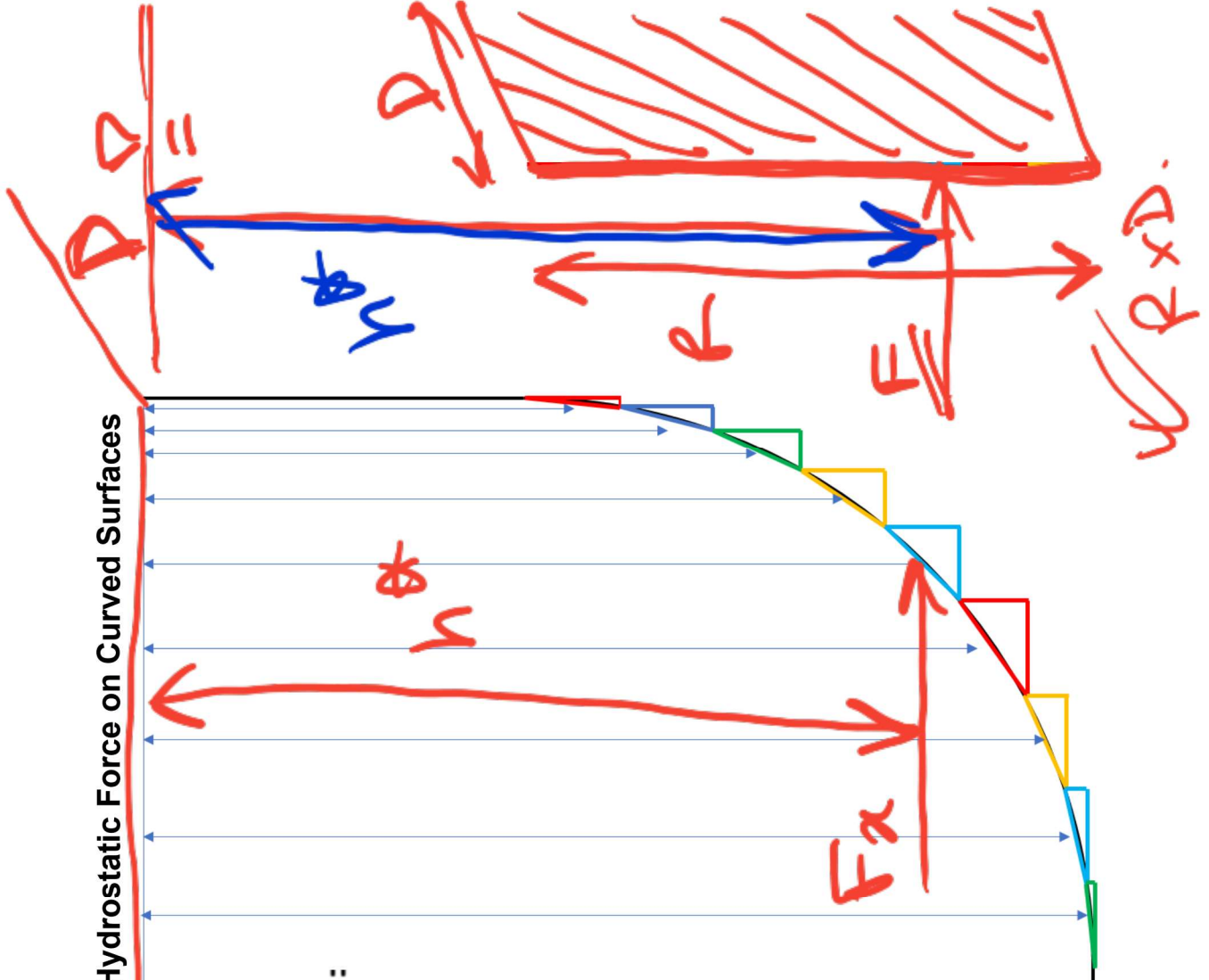
$$F_y = \int dF_y$$

# Horizontal Component of Hydrostatic Force on Curved Surfaces

$$F_x = \int dF_x = \int \rho g h dA \sin \theta$$



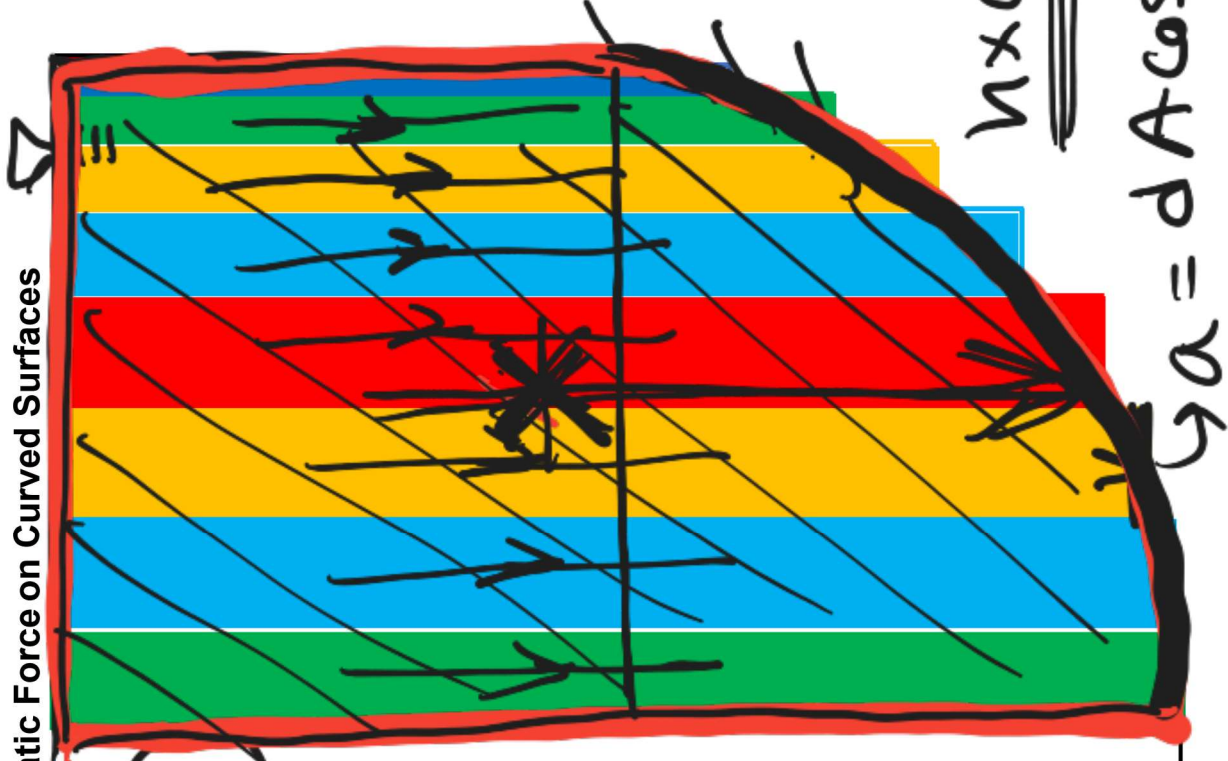
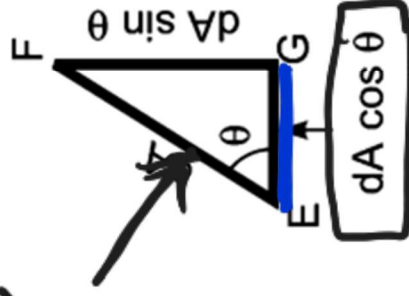
$$\rho g h dA \sin \theta$$



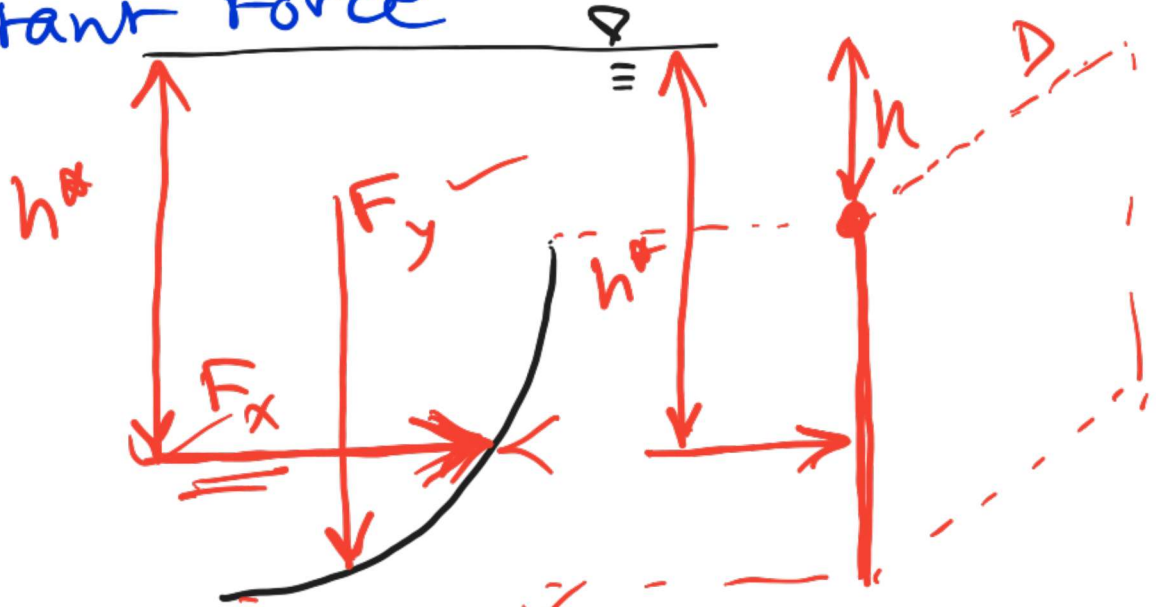
# Vertical Component of Hydrostatic Force on Curved Surfaces

$$F_y = \int dF_y = \int \rho g h dA \cos \theta$$

$$\rho g \underbrace{h dA \cos \theta}_{\substack{= \text{Weight} \\ \text{of fluid above}}}$$



# Magnitude and Direction of Resultant Force

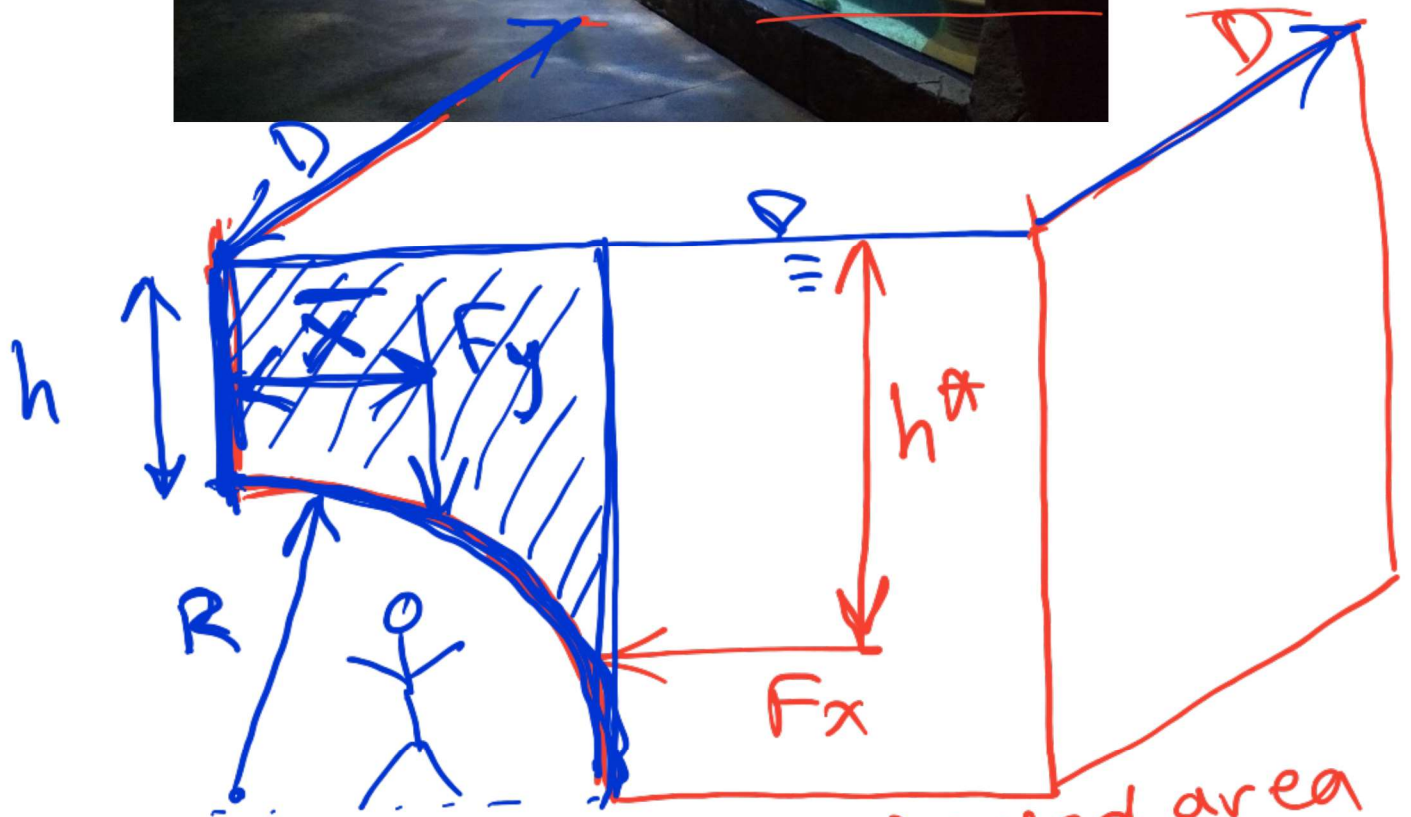
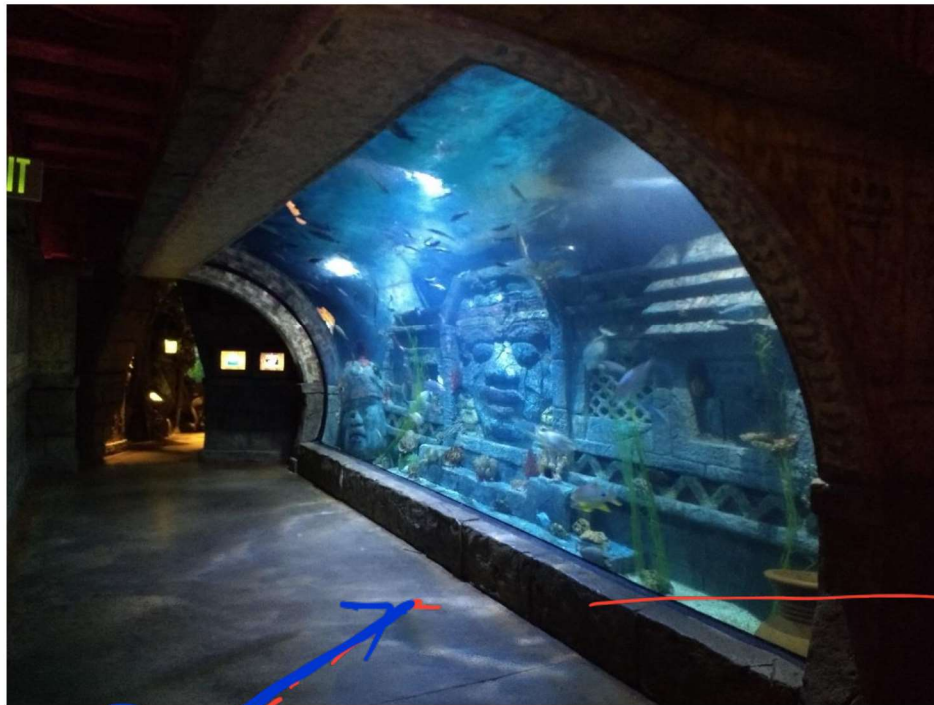


$$F_R = \sqrt{F_x^2 + F_y^2}$$

$$\tan \theta = \frac{F_y}{F_x}$$

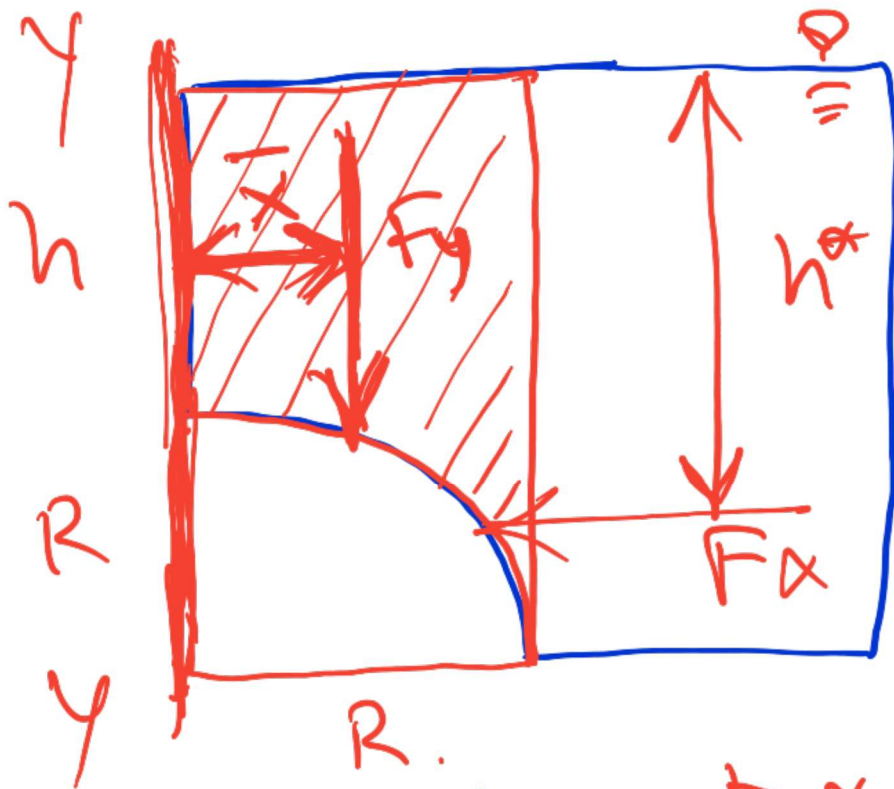
$$\theta = \tan^{-1} \left( \frac{F_y}{F_x} \right)$$





$F_x$   $\Rightarrow$  Force on projected area  $R \times D$

$F_y$   $\Rightarrow$  weight of liquid above curved surface.

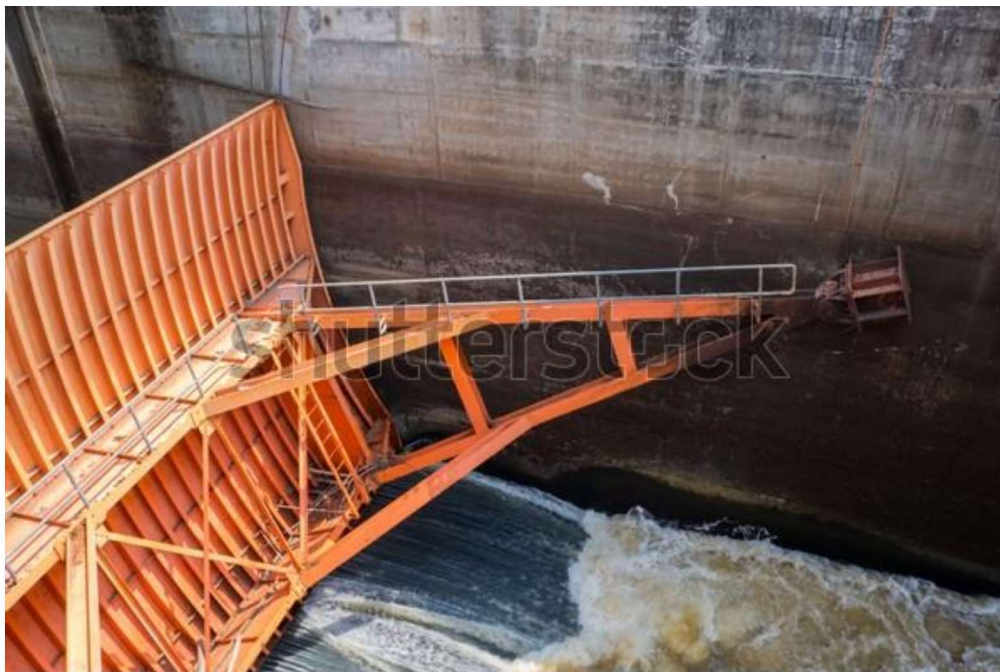
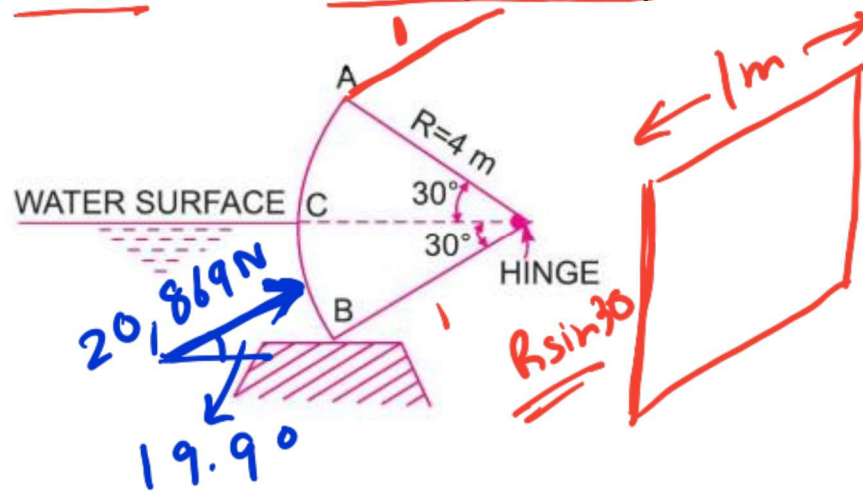


Centroid of rect  $\times$  area  
of rect

$= (\text{Centroid of shaded area})$   
 $\times$  area of shaded  
area

$+ (\text{Centroid of quarter}$   
 $\text{circle}) \times$  area of  
quarter  
circle

Find the horizontal and vertical components of the water pressure exerted on a radial gate of radius 4 m. Consider width of gate as unity.



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<https://www.shutterstock.com/image-photo/radial-arm-gate-used-dam-control-1057660106>



$$F_x = \bar{h} \rho g A$$

$$= 1 \times 1000 \times 9.81$$

$$\times 2 \times 1$$

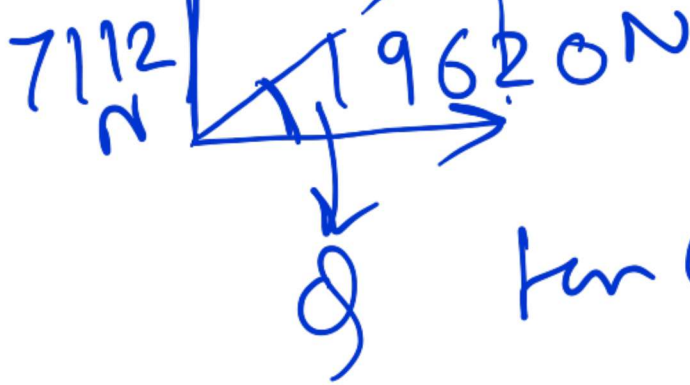
$$= 19620 \text{ N}$$

$$F_y = \left( \frac{30}{360} \times \pi \times 4^2 - \frac{1}{2} \times 4 \cos 30^\circ \times 4 \sin 30^\circ \right) \times 1 \times 9810$$

$$= 7112.25 \text{ N}$$

$$F_R = \sqrt{F_x^2 + F_y^2} = \sqrt{19620^2 + 7112^2}$$

$$= 20,869 \text{ N}$$



$$\tan \theta = \frac{7112}{19620}$$

$$\theta = \underline{\underline{19.9^\circ}}$$